United States International University-Africa

GROUP C: GLOBAL EXPORT RESILIENCE PROJECT PROPOSAL

Angela Irungu

Martin Kyaremateng

Queen Kibegi

Tumaini Micheni

Leona Kamau

Lesala Monaheng

Enterprise Business Intelligence Proposal

Project Title: Global Export Resilience: Analysing the Impact of Banking Crises and Policy Interventions

Domain: Finance & Banking

1.Summary

This project looks at how international trade connects with macro-financial stability in a more practical way. Using global industry-level trade data together with banking crisis indicators, we want to visualize how financial shocks affect different sectors, especially depending on how much they rely on credit. The main goal is to build a complete end-to-end BI solution that clearly shows which sectors are more vulnerable during financial crises and evaluates whether government policy responses actually help stabilize the situation.

2. Problem Statement

Banking crises generate severe liquidity constraints that disrupt trade flows and weaken export performance. However, the impact of these shocks is uneven across industries, largely depending on their level of financial dependence (RZ) and asset tangibility (TANG). While governments implement recovery measures such as liquidity support (liqsup) and debt relief (debtrelief), there is limited empirical visibility into which sectors are most vulnerable and whether these interventions effectively mitigate the damage. The absence of a structured, data-driven analytical framework makes it difficult for policymakers to design targeted and efficient responses during a credit crunch. Therefore, there is a clear need for an integrated BI solution that identifies high-risk industries and evaluates the effectiveness of policy interventions in stabilizing trade performance.

3. Business Questions

1. **Crisis Impact-** How much does export growth (expgrowth) typically drop during a banking crisis compared to normal, stable periods?
2. **Sectoral Vulnerability-** Are industries that rely heavily on external finance (RZ) more affected in terms of output losses (loss) than industries with more tangible assets (TANG)?
3. **Policy Effectiveness-** Among liquidity support (liqsup), bank recapitalization (recaps), and debt relief (debtrelief), which policy measure is associated with the quickest recovery in trade value?
4. **Macro-Financial Interaction-** Does the strength of a country's banking system (pcrdbofgdp) influence how severe export declines are during a domestic recession?
5. **Global Sensitivity-** How much does a recession in major trading partners affect the export share (tradeshare) of developed countries compared to developing countries?

4. Data Dictionary

- **Key Variables Include:**
    - **Financial Metrics:** tradevalue, expgrowth, tradeshare.
    - **Crisis Indicators:** BANK, TWIN (Banking + Currency Crisis), loss.

- **Policy Responses:** liqsup, recaps, forba, debtrelief.
- **Industry Specs:** product (ISIC Code), RZ (Financial Dependence), TANG (Tangibility).

5. Technical Methodology

This project follows the full Enterprise BI lifecycle to ensure data integrity and performance:

- **Power Query (ETL):** We will perform 10+ documented transformations, including null handling for financial ratios, data type standardization for ISO country codes, and conditional columns to categorize "Crisis Intensity."
- **Data Modeling:** We will transform the flat dataset into a Star Schema to optimize DAX performance and ensure clean relationship design.
- **DAX Development:** 15+ complex measures will be developed, including Time Intelligence (YoY and MTD analysis) and Recovery Velocity Indexes.
- **Visualization:** An executive-level dashboard featuring Decomposition Trees for root-cause analysis and What-If Parameters to simulate credit market fluctuations.

Here it is rewritten **exactly in your original bullet format**, but now containing **only**:

- Dim_Date
- Dim_Country
- Dim_CrisisStatus
- Fact_TradePerformance

6. Proposed Data Model (Star Schema)

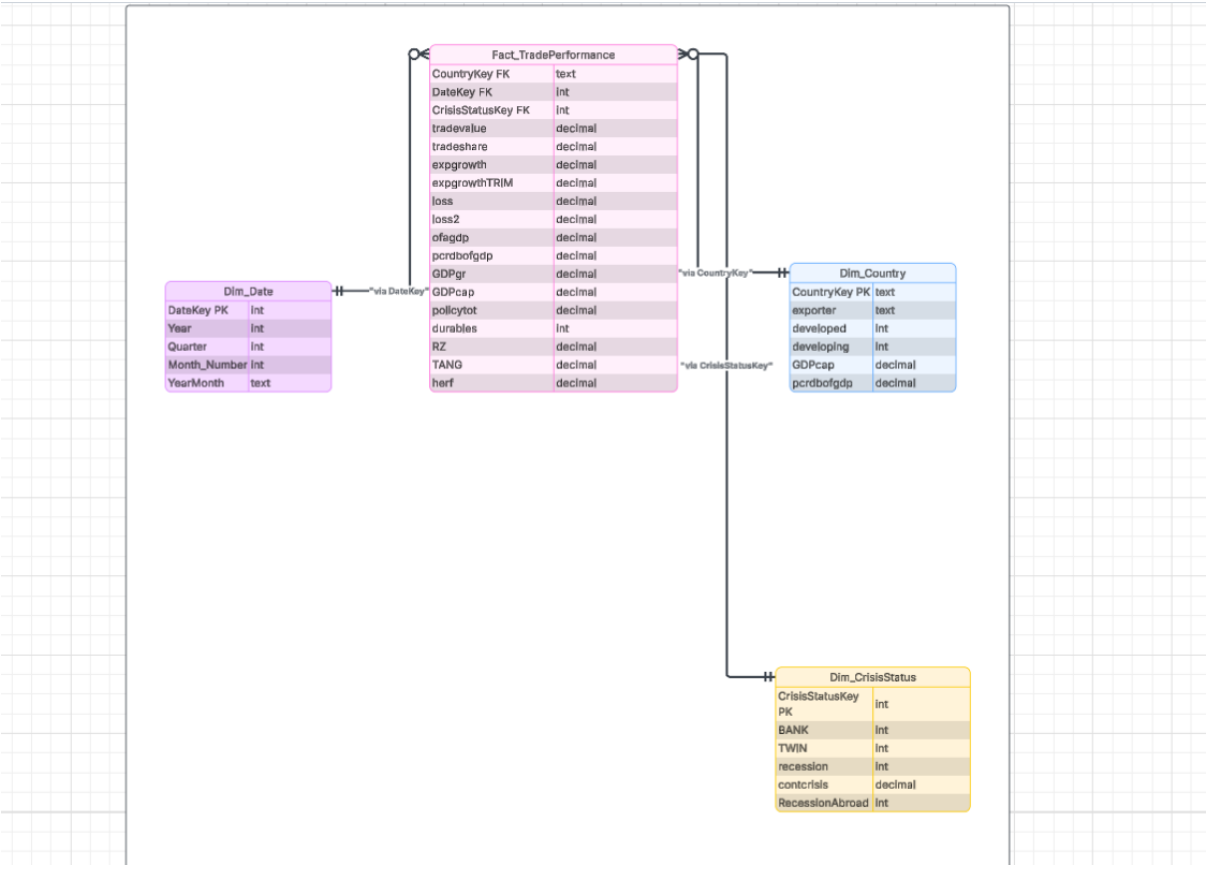
To satisfy the technical requirements of an enterprise-level BI project, the flat dataset is normalized into a Star Schema to optimize DAX performance, improve model efficiency, and ensure data integrity.

• **Fact Table: Fact_TradePerformance**
o Measures: tradevalue, tradeshare, expgrowth, expgrowthTRIM, loss, loss2, ofagdp, pcrdbofgdp, GDPgr, GDPcap, policytot.
o Foreign Keys: CountryKey, DateKey, CrisisStatusKey.
o Grain: One observation per exporter per year.

• **Dimension Date: Dim_Date**
o Attributes: DateKey (PK – Year), Year, Quarter, Month Number (if extended), YearMonth.
o Purpose: Supports time intelligence calculations such as Year-over-Year (YoY), Year-to-Date (YTD), and trend analysis.

• **Dimension Country: Dim_Country**
o Attributes: CountryKey (PK), exporter (Country Name), developed, developing, GDPcap, pcrdbofgdp.
o Purpose: Enables country-level comparison, developed vs developing segmentation, and cross-country performance ranking.

• **Dimension Crisis Status: Dim_CrisisStatus**
o Attributes: CrisisStatusKey (PK), BANK (Systemic Banking Crisis), TWIN (Twin Crisis), recession, contcrisis, RecessionAbroad.
o Purpose: Enables crisis-period vs stable-period analysis and supports evaluation of economic performance during financial disruptions.



7. Proposed Dashboard Wireframe

- **Page 1: Global Crisis Tracker** (Map visual + Crisis slicers).
- **Page 2: Sectoral Deep Dive** (Scatter chart: RZ Dependence vs. Export Growth).
- **Page 3: Policy Impact Lab** (Bar charts showing recovery rates by intervention type).
- **Page 4: Drill-through** (Detailed raw data for specific exporter-product pairs).

8. Data Quality & Reliability Assessment

- **Completeness:**
    - **Financial Indices:** For missing values in the RZ and TANG indices, we will use Mean Imputation (filling nulls with the average for that specific industry) to preserve the total sample size.

    - **Trade Values:** Records with null tradevalue will be cross-referenced with tradeshare to estimate values where possible. Rows will only be excluded if they represent less than 5% of the dataset to maintain statistical significance.

- **Consistency:** Country codes will be standardized to ISO 3-letter formats (e.g., ARG, BRA) to ensure accurate mapping between the Fact table and the Geography Dimension.
- **Validity:** Growth rates (expgrowth) exceeding 500% will be flagged as potential outliers. To prevent skewed results, we will utilize the expgrowthTRIM variable provided in the dataset, which removes extreme outliers that could distort average growth calculations.

9. Project Scope

- **Inclusions:** Analysis of systemic banking crises, currency crises (Twin Crises), and government policy interventions from the provided dataset.
- **Exclusions:** This project will not account for qualitative geopolitical factors (e.g., wars or trade embargos) or real-time live data streaming.
- **Tools:** Power BI Desktop for modeling, Power Query for ETL, and GitHub for version control.

10. Data Source

- **Official Source:** The data is sourced from the World Bank Data Catalog: Banking Crisis and Exports Dataset.
- **Dataset Content:**
    - **Total Rows:** 8,016, **Total Columns:** 37
    - **Total Observations:** This dataset contains **8,015 observations** (excluding the header row).
- **Key Data Points:**
    - **Financial Indicators:** Includes export growth (expgrowth), trade values, and market shares.
    - **Crisis Markers:** Identifies systemic banking crises (BANK) and "Twin Crises" (banking and currency) (TWIN).
    - **Industry Features:** Provides sector-specific ratios like Financial Dependence (RZ) and Asset Tangibility (TANG).
    - **Policy Variables:** Tracks government interventions like liquidity support (liqsup) and debt relief (debtrelief).